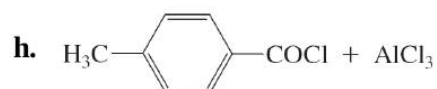
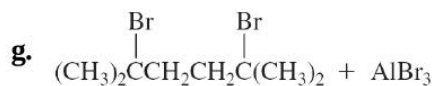
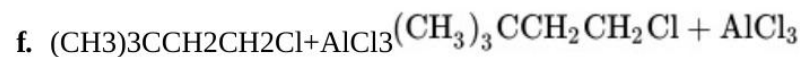
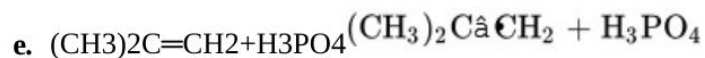
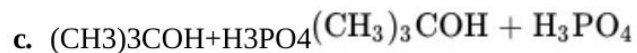
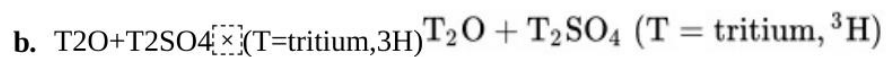


47. Reaction review. Without consulting the Reaction Road Map on [p. 749](#),

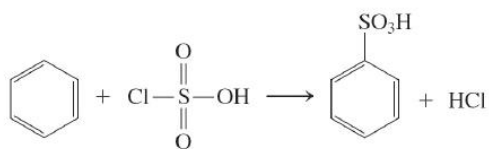
48. Give the expected major product of addition of each of the following reagent mixtures to benzene. (**Hint:** Look for analogies to the reactions presented in this chapter.)



49. Write mechanisms for the reactions of parts (c) and (f) in [Problem 48](#).

50. Hexadeuteriobenzene, C_6D_6 , is a very useful solvent for ^1H NMR spectroscopy because it dissolves a wide variety of organic compounds and, being aromatic, is very stable. Suggest a method for the preparation of C_6D_6 .

51. Propose a mechanism for the sulfonation of benzene using chlorosulfuric acid, ClSO_3H (given below).

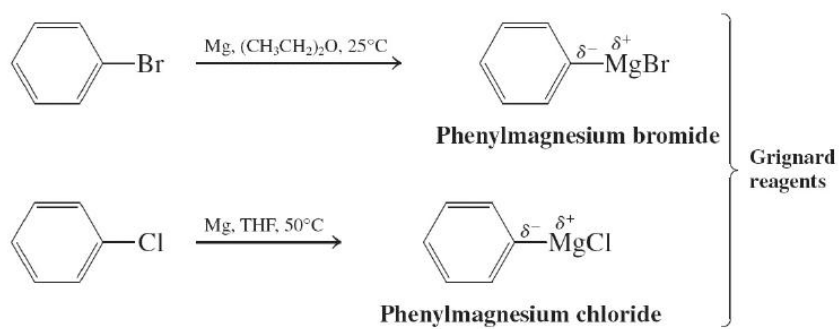


52. Benzene reacts with sulfur dichloride, SCl_2 , in the presence of AlCl_3 to give phenylthiobenzene, $\text{C}_6\text{H}_5\text{-S-C}_6\text{H}_5$. Propose a mechanism for this process.

53. (a) 3-Phenylpropanoyl chloride, $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{COCl}$, reacts with AlCl_3 to give a single product with the formula $\text{C}_9\text{H}_8\text{O}$ and an ^1H NMR spectrum with signals at $\delta = 2.53$ (t, $J = 8$ Hz, 2 H), $\delta = 3.02$ (t, $J = 8$ Hz, 2 H), and $\delta = 7.2-7.7$ (m, 4 H) ppm. Propose a structure and a mechanism for the formation of this derivative.
- (b) The product of the process described in (a) is subjected to the following reaction sequence: (1) NaBH_4 , $\text{CH}_3\text{CH}_2\text{OH}$; (2) conc. H_2SO_4 , 100°C ; (3) H_2 , Pd-C, $\text{CH}_3\text{CH}_2\text{OH}$. The resulting molecule exhibits five resonance lines in its ^{13}C NMR spectrum. What is the structure of the substance formed after each of the steps in this sequence?

54. The text states that alkylated benzenes are more susceptible to electrophilic attack than is benzene itself. Draw a graph like that in [Figure 15-20](#) to show how the energy profile of electrophilic substitution of methylbenzene (toluene) would differ quantitatively from that of benzene.

55. Like haloalkanes, haloarenes are readily converted into organometallic reagents, which are sources of nucleophilic carbon.

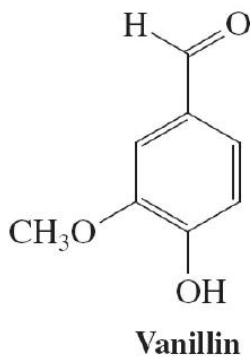


56. Give efficient syntheses of the following molecules, beginning with benzene. **(a)** 1-Phenyl-1-heptanol; **(b)** 2-phenyl-2-butanol; **(c)** octylbenzene. **(Hint:** Use a method from [Section 15-13](#). Why will Friedel-Crafts alkylation not work?)

57. Vanillin is a benzene derivative with several functional groups, each one of which displays its characteristic reactivity. What would you expect to be the products of reaction of vanillin with each of the following reagents?

a. $\text{NaBH}_4, \text{CH}_3\text{CH}_2\text{OH}$ $\text{NaBH}_4, \text{CH}_3\text{CH}_2\text{OH}$

b. NaOH , NaOH , then CH_3I CH_3I



Vanillin is obtained by extraction from the seed pods of plants in the *Vanilla*

43. Give the expected major product(s) of each of the following reactions.

52. Write the expected product(s) of each of the following reactions.

